Research Brief: Episodic & Vector-Space Memory for Language Models

Prepared for: Goals demo ? research brief ? verified findings
Date: 2026-10-26
Scope length: ~1?2 printed pages
Confidentiality: Internal demo fixture (no secrets)

Background

Long-context models still forget, contradict, and mix sessions. Practitioners therefore bolt on **external memory**: write past turns or tool results into a store, retrieve top?k chunks by embedding similarity, and recombine them into the next prompt.

Three ideas keep showing up under different names:

1. **Episodic memory** ? store discrete past events (dialog turns, tool traces, ?what happened?) rather than only parametric weights.
2. **Vector-space memory** ? encode those events as dense vectors; retrieve by nearest-neighbor search (FAISS, HNSW, etc.).
3. **Context recombination** ? stitch retrieved snippets (and sometimes summaries) back into the model?s working context under a token budget.

This brief asks an agent to **locate primary sources**, extract **verbatim claims**, and separate marketing from peer-reviewed or open-source evidence. Do **not** invent DOIs, arXiv IDs, or GitHub stars. Prefer sources the agent can open and quote.

Topic targets the agent must find and verify

Treat these as **search targets**, not as already-verified citations:

Target (search for)	Why it matters
MemGPT / Letta ? LLM OS / virtual context management	Episodic-style paging of memory in/out of context
Generative Agents (Park et al.) ? memory stream + retrieval	Classic episodic memory for simulated agents
RETRO / RETRO++ (DeepMind) ? retrieval-augmented generation at scale	Vector retrieval as part of generation
HippoRAG / HippoRAG 2 ? hippocampal-inspired RAG	Graph + embedding recombination claims
LangChain / LlamaIndex memory modules; Hugging Face datasets tagged ‘retrieval‘ or ‘memory‘	Implementation reality vs paper claims

Confirm each hit with a **primary URL** (arXiv abs/pdf, HF model/dataset card, or GitHub README). Discard secondary blog posts that lack a link to a primary artifact.

Research questions

Answer each with evidence. Prefer short answers plus **verbatim quotes** (see Success criteria).

1. **Write path:** How do published systems decide *what* to write into episodic or vector memory (full turns, summaries, embeddings of both)? Cite at least one paper and

one open-source README.
2. **Retrieve path:** What similarity metric and index type do the most-cited open implementations use by default (cosine vs L2; FAISS Flat/IVF/HNSW; other)? Quote the README or paper method section.
3. **Recombination:** Under a fixed context window, what policies appear for merging retrieved chunks with the live dialogue (prepend, interleave, summarize-then-inject, hierarchical)? Name two distinct policies with sources.
4. **Failure modes:** Which documented failure modes are attributed to memory stores rather than the base LM (stale facts, cross-session leakage, embedding drift, ?lost in the middle?)? One quote per failure mode, max three modes.
5. **Eval:** What datasets or benchmarks are used to measure long-horizon memory (not just single-hop RAG)? Prefer HF dataset cards or arXiv eval sections; list metric names exactly as written.
6. **(Optional stretch)** Is there a public **model card** that claims built-in episodic memory (vs external RAG)? If none found after a good-faith search, state that explicitly with query terms used.

Preferred source types

Use these in priority order:

1. **arXiv** abs pages and PDFs (method + experiments sections).
2. **Hugging Face** model cards and dataset cards (limitations / evaluation blocks).
3. **GitHub READMEs** and linked design docs in the same repo (version or commit pinned if possible).
4. Peer-reviewed PDFs only when freely reachable; do not rely on paywalled abstracts alone.

Avoid: anonymous Medium posts, SEO roundups, and ?10 best memory libraries? listicles **unless** they deep-link to a primary source you then quote.

Success criteria

A finding counts as **verified** only if all of the following hold:

- **Named artifact:** title + authors or org + stable URL.
- **Verbatim quote:** ?1 sentence copied exactly from the primary source, in quotation marks, with section name or page if available.
- **Claim ? quote link:** each answer to a research question maps to at least one quote; no unsupported paraphrase.
- **Contradiction check:** if two sources disagree (e.g. default metric), report both quotes and label the conflict.
- **No fabricated identifiers:** never invent arXiv IDs, DOIs, commit hashes, or star counts. If uncertain, write ‘UNVERIFIED‘ and the search query.

Deliverable shape for the Goals agent: a short findings memo with one subsection per research question, each ending in ‘Quote:‘ / ‘Source:‘.

Out of scope

- Training new embedding models or fine-tuning LMs.

- Production architecture for Theorem / this repo?s guardrail stack.
- Commercial vendor whitepapers without a public technical artifact.
- Speculative neuroscience analogies unless the paper itself uses them and you quote that usage.
- API keys, private Hub tokens, or scraping behind login.
- Claiming ?SOTA? without the paper?s own metric table quoted.

Constraints for the demo agent

- Stay within publicly readable sources.
- Budget: enough depth for **four** of the six questions with full verification; mark the rest ‘PARTIAL‘.
- If attachment text is truncated, prioritize questions 1, 3, and 5.
- Do not modify files outside the findings output the host application designates.